

Department of Computer Science and Engineering
Academic Year 2021– 2022(Even Semester)

Degree, Semester & Branch: B.E, IV & CSE

Course Code & Title: CS8394 & Operating System

Name of the Faculty member (s): Dr. I Gethzi Ahila Poornima

Innovative Practice Description

- **Unit / Topic:** Unit IV/ Disk Scheduling Algorithms
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO 2
- **Activity Chosen:** Jigsaw-Collaborative Learning
- **Justification:**

Jigsaw is a cooperative group activity with a twist: under the teacher's supervision, students efficiently teach one another. The way that students engage with one another influences what they learn about a certain topic, skill, or problem.

The topic picked reveals the disc request scheduling order. The amount of disc arm movement or delay is calculated using these algorithms. This topic is better suited for a Jig-Saw cooperative learning exercise where students implement the algorithms and then examine the optimal disc scheduling algorithm to minimize needless disc movement and delay. Additionally, questions of this nature are commonly posed in GATE and other competitive exams for jobs in the public sector. All of the disc scheduling algorithms are learnable by students. Through this activity, students will be able to improve their peer learning, self-learning, individual and teamwork, and communication skills.

- **Time Allotted for the Activity:** 40 Minutes
- **Details of implementation**

Three/Four-person diverse groups were created. Depending on their academic achievement such as CGPA of the previous end semester or marks scored in previous Internal Assessment Test, students are categorized as bright, average, or slow learners. Each group comprises of at least one person from each category. We had 21 groups in the class.

A leader is chosen from each group. Each member in the group is assigned a number from 1 to 3.

- 1- FCFS Disk Scheduling
- 2- Shortest Seek Time First
- 3- SCAN and LOOK

Students from home groups who received the same number were divided into expert groups. Each expert group member had 15 minutes to debate a predetermined subtopic. The expert group of students returned to their home groups and were given 10 minutes for additional conversation, during which time each member became knowledgeable about the disk scheduling algorithms. The expert group students brought back into their home groups and allotted 10 minutes for further discussion in which each member become expert in applying disk scheduling algorithms to calculate the disk movement made to

satisfy all the disk requests. 10 minutes were allotted for the presentation. One student from each home group presented his learned content. The others in the group were encouraged to raise queries for clarification in various disk scheduling algorithms. A few sample snaps during this event is shown in Figure 1.

- **CO – PO / PSO mapping specific to this event:**

CO	PO1	PO4	PSO1
CO5	1	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped Justification:**

Innovative practice Jigsaw	PO1	PO4	PSO1
	1	2	2
Justification for correlation	Students could apply the disk scheduling algorithms to analyze the performance in terms of disk arm movement and delay.	Students will be able to conclude which disk scheduling can be followed to improve the throughput of disk response.	The basic understanding of the disk scheduling algorithms will make the students to excel in choosing the suitable disk scheduling algorithms in designing Operating systems.

- **Images / Screenshot of the practice:**



Figure 1: Sample sheet & picture taken

- **Reflective Critique:**
 - ❖ ***Feedback of practice from students and other stakeholders:***
 - Most of the students felt that this event helped them to calculate the disk movements and delay for the given IO requests.
 - ❖ ***Benefit of the practice:***
 - All the students enthusiastically involved in this discussion activity.
 - They felt that this event encourages them to develop self and peer learning.
 - ❖ ***Challenges faced in implementation:***
 - Few students hesitated to take part in expert group discussions.

Signature of Faculty Member

HOD